8D report

Customer plant	Eurotec	**Supplier**	AVO Carbon Germany	**Date of claim**	19.02.2024
Customer contact	A. Strnatkova	**AVO contact**	J.Gehl		
Customer reference	24008	**AVO reference**	240202		

QUICK RESPONSE (D1-D3)
This document must be issued within 24 hours after claim notification

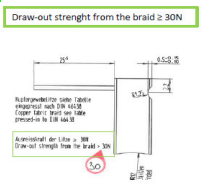
D 1. Description of the problem

1.a What is the problem ? Use 5W+2H to describe what is the current situation.

Problem description	Draw-outstrenght from the braid≥ 30N - measured 19,3N, 22N, 23,8N Cracked/broken edge on the carbonbrush close to wire area	What happened?	Draw-outstrenght from the braid≥ 30N - measured 19,3N, 22N, 23,8N Cracked/broken edge on the carbonbrush close to wire area
Incident type	Customer claim	Why is it a problem?	Parts can not be assembled.Out of spec.
PN customer/AVO	27579.52267 / 1020510	When did it happen?	
Part description	Carbon brush	Who detected it?	No information
AVO batch number	277910	Where was it detected?	No information
Recurrence	non reccurent	How was it detected?	No information
Defect S/R	no	How many nOK parts ?	14
Standard process used	yes		

nok part picture / sketch

ok part picture / sketch



1.b Problem solving team

Name	Position	Location	Email
Joachim Gehl	QM	AVO Germany	joachim.gehl@avocarbon.com
Martin Röhr	PRD	AVO Germany	
Mustafa Yorulmaz	KVP	AVO Germany	
Joachim Gehl	QM	AVO Germany	
Giovanni Balenzano	PRD	AVO Germany	

1.c Tools to use

yes	**Ishikawa**	yes	**5why's**		**LLC**		**Audit**		Other

1.d First analyse of problem / potential root causes

Potential root cause(s): Please send us the defective parts for analysis.

The samples are mandatory for the deeper analysis. Feel free to use our FedEx account. (Please see below marked in yellow)

Samples of nok parts required in plant for analyse: yes

Express shipping number account: FedEx 186141300

D 2. Risk on similar products and processes

Can this defect happen on similar parts ?

Other family product	---
Other line	---
Other tools/work station	---
Other	---

Comments (compulsory if yes)

D 3. Containment

Sorting criterias (add additional sheets if needed)

Location in the supply chain	Nb of involved parts	Nb parts sorted	Nb nok parts	ppm	Date	Resp.
Parts at customer	**30.461**	---	**14**	---	19.02.2024	A. Strnatkova
Parts in transport		---	---	---	---	---
Parts at AVO Germany stock	**0**	---	---	---	20.02.2024	G. Luschtinetz
Parts at AVO production	---	---	---	---	---	---

First contained shipment information

Immediate actions to prevent production and delivery of nok parts ?

Actions	Resp.	Date plan.	Date clos.
Register Complaint	G. Luschtinetz	20.02.2024	20.02.2024
Information of production	G. Luschtinetz	20.02.2024	20.02.2024
Quality alert	G. Luschtinetz	20.02.2024	20.02.2024
Check the production-, quality- and maintenance data	G. Luschtinetz	20.02.2024	20.02.2024
Sorting parts at customer site with microscope (estimated sorting costs 1270 €)	A. Strnatkova	20.02.2024	20.02.2024

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PLAN - DO (D4-D6)

D 4. Root cause of occurrence

Work out the root causes using Ishikawa and 5 why's

Root cause of occurrence :

The holes in the upper punch for guiding the wires, were made at the lower tolerance limit. This means that the stranded wire is sometimes compressed at the press-fit point, which damages the material. Cutting with scissors creates pull force on the wire and further damages the carbon brush.

How was the defect reproduced ?

With the causing punch. Holes are on the lower tolerance limit.

D 5. Root cause of non detection

Work out the root causes using Ishikawa and 5 why's

Root cause of non detection :

The hourly inspection could not detect the parts due to the insufficient sample size. This is related on the fact that the punch holes are in a technically proper condition.

D 6. Corrective actions and non detection

Actions	Resp.	Date plan.	Date clos.
Reduce the lower tolerance of the punches	U. Wallocha	13.03.2024	13.03.2024
Test: 180° turned pressing tool to reduce the pull force on the wire	M. Yorulmaz	tbd	15.03.2024
Increase sample check size in the final inspection	M. Altun	13.03.2024	13.03.2024

CHECK - ACT (D7-D8)

D 7. Effectiveness of action plan

Actions to check effectiveness (to plan with part D6)	Resp.	Date plan.	Date clos.
Sorting after all implementation	Sorting team	tbd	

D 8. Lessons learned

8.1 Do following documents require an update ?

	Resp.	Date plan.	Date clos.
PFMEA	Mr. OTT	21.03.2024	
		verified	changed
			yes / no
Control Plan	Mr. OTT		
		verified	changed
			yes / no
Work instruction			
Control instruction			
Procedure			
Control gage			
Maintenance instruction			
Lesson Learned Card			
Other			

8.2 Generalization of countermeasures to similar lines/ products

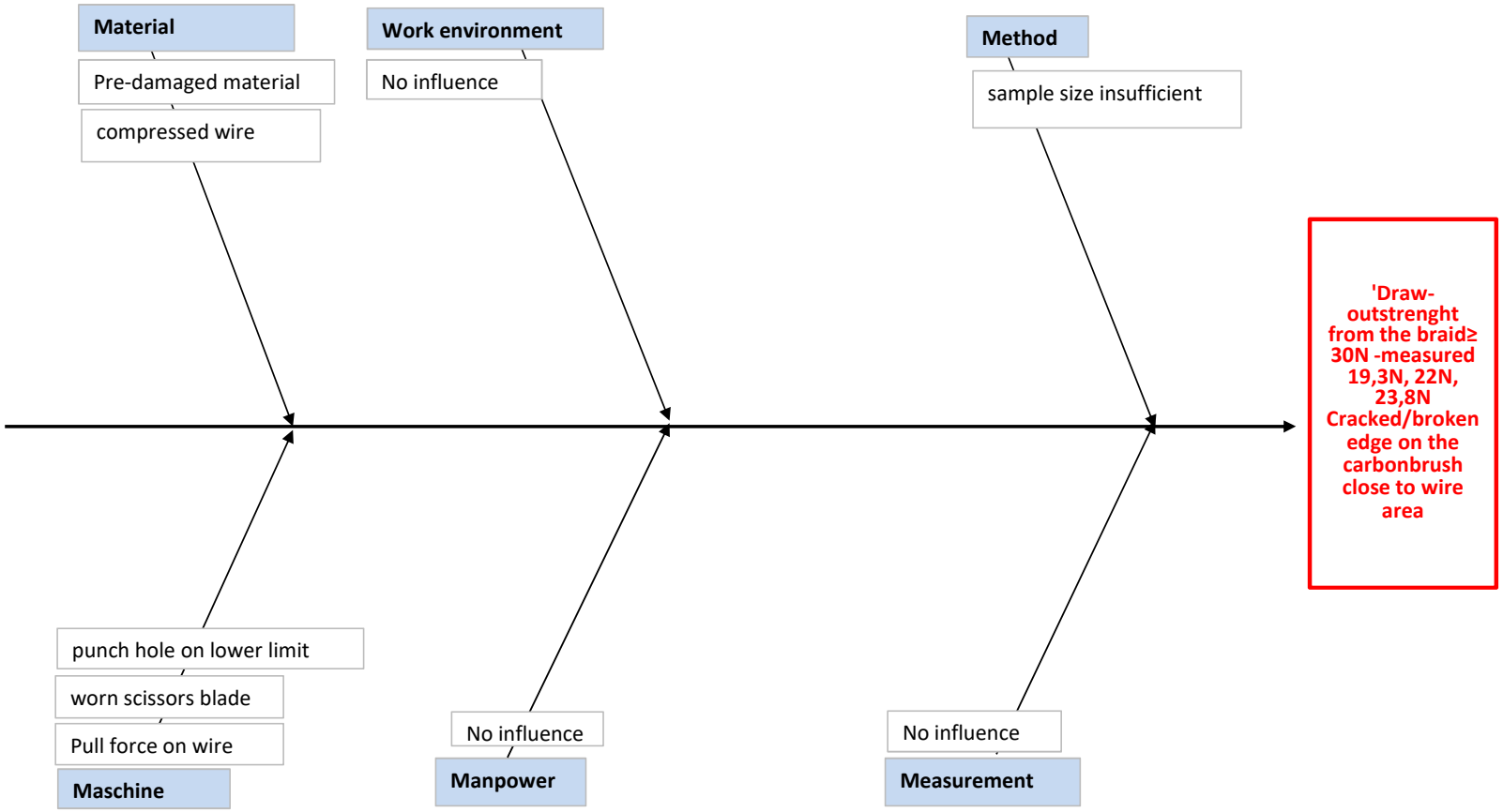
8.3 Internal audit (if required)

Date

Results

Comments

Ishikawa


Material
Pre-damaged material
compressed wire
Work environment
No influence
Method
sample size insufficient
'Draw-outstrenght from the braid≥ 30N -measured 19,3N, 22N, 23,8N Cracked/broken edge on the carbonbrush close to wire area
punch hole on lower limit
worn scissors blade
Pull force on wire
Maschine
No influence
Manpower
No influence
Measurement

5 Why's

Problem description : **Draw-outstrenght from the braid≥ 30N -measured 19,3N, 22N, 23,8N Cracked/broken edge on the carbonbrush close to wire area**

Problem statement	Nb.	1st WHY?	1st ANSWER	2nd WHY ?	2nd ANSWER	3rd WHY ?	3rd ANSWER	4th WHY ?	4th ANSWER	5th WHY ?	5th ANSWER
OCCURENCE											
	1	Why is the brush damaged?	Because the wire is not straigt pressed in to the brush.	Why is the wire not straigt pressed in to the brush?	Because the wire is compressed.	Why is the wire compressed?	Because the punch holes to small.	Why are the punch holes to small?	Because punch hole was machined on the lower tolerance.		
NON DETECTION											
	1	Why is the low pull out force not detected?	Because only a smallsample size is checked hourly.	Why is only a small sample size checked hourly?	Because it is the sample size frequence for checking.						